

Designing News Feed Service →

- ques → What type of requests do we need to support?
- ques → What type of content does a message can hold?
- ques → What are the number of users?
- ques → How long do we need to retain the posts in the DB for an user?
- ques → How do we display the feed?

Starting with the first question let say for now, we only hold 3 type of request →

- Post Request → this will post the content after authentication.
- News Feed → this will get all the post of friends from our friends list.
- update Request → let say if we need to add some friend, this will add the friend in the db list.

• Now coming to the second question, let say a post can contain image, video and text as well.

• let say there are total 10M number of DAU.

• total number of DAU → 10×10^6

Number of post a user can make in a day → 5

Average size of post a user can post → 1MB or 100KB

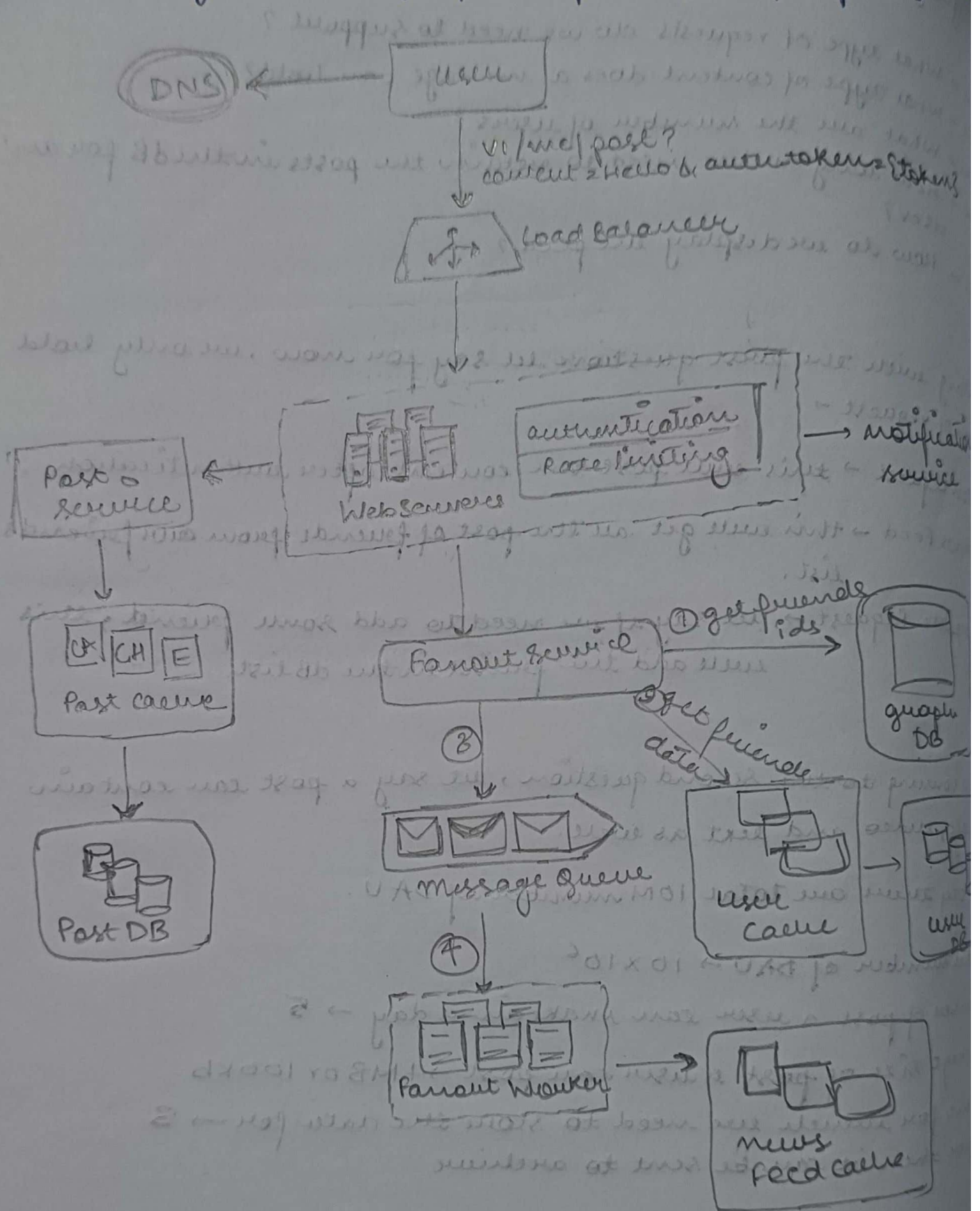
Time for which we need to store the data for → 3
after then it can be sent to archive

Total number of post in a day → 5×10^7

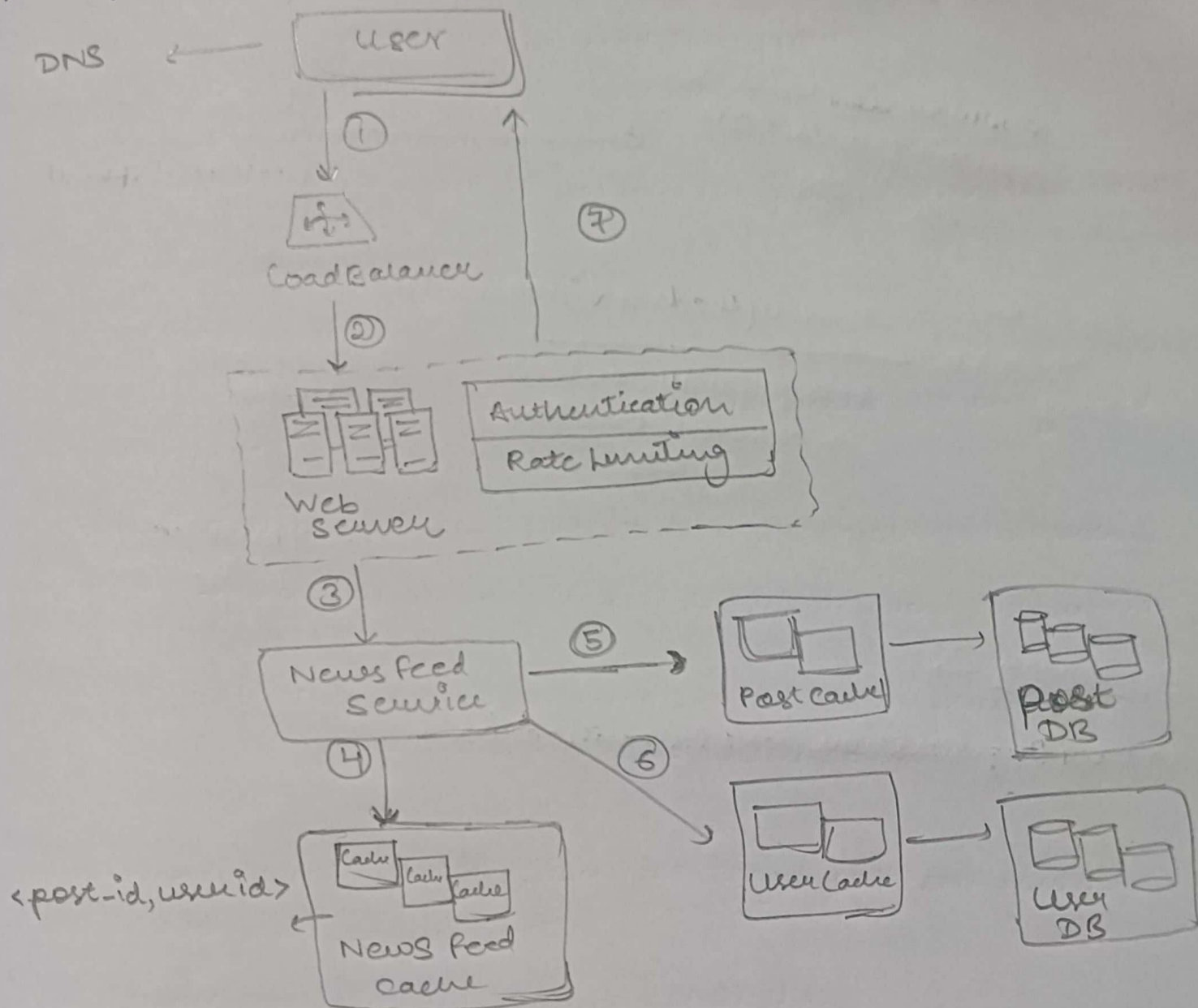
Total amount of memory used in a day → $100 \times 10^3 \times 5 \times 10^7$
→ 5×10^{12}
→ 5TB.

• Let say we display the feed on the reverse chronological order.

Let make a high level design for the first type of request



News feed retrieval high level design →



list of cache →

(a) News Feed Cache → news feed.

(b) Content → Hot Content, Normal Content

(c) Social Graph → follower, following

(d) Action → liked, replied, other

(e) counter → like counter, reply counter.